

Crowded Carry in G10 FX and Crypto

Crash risk in FX and crypto carry trades

STUDENTS

Reza Zamani · Paraj Goyal

PROFESSOR

Amir Kermani

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Carry pays because it crashes

Our question: do carry trades crash when they are crowded?

- We test it in two markets with the same economics: G10 currencies and crypto.
- Strategy A is carry in seven G10 currencies against the US dollar, with a crowding filter built from CFTC data.
- Strategy C is funding carry in twelve crypto perpetual futures on Binance, including the coins that collapsed, Terra and FTX Token.

Why carry crashes

- Carry borrows the low-yield asset and holds the high-yield one. UIP says this earns nothing, but in the data it earns a premium, with rare large crashes.
- Crashes are worst when the trade is crowded and everyone unwinds at once (Brunnermeier, Nagel & Pedersen 2008).

Hypotheses, fixed before testing

- A: carry earns a premium, crowded months are followed by weaker carry, and it loses in volatility spikes.
- C: funding carry earns a premium with little BTC exposure, and its crashes are coin-specific.

A: G10 FX

C: Crypto

Carry signal	3-month rate differential	Perpetual funding rate
Crowding data	CFTC speculator positions	Funding = leveraged demand
Universe	7 currencies vs USD	12 coins (listed below)
Frequency	Monthly, 1999–2026	Weekly, 2020–2026

Currencies: Japanese yen (JPY), euro (EUR), British pound (GBP), Swiss franc (CHF), Canadian dollar (CAD), Australian dollar (AUD) and New Zealand dollar (NZD), each against the US dollar.

Coins: Bitcoin (BTC), Ethereum (ETH), BNB, XRP, Cardano (ADA), Dogecoin (DOGE), Solana (SOL), Litecoin (LTC), Chainlink (LINK), Avalanche (AVAX), Terra (LUNA) and FTX Token (FTT).

Data and methodology

Public data; a research design chosen to keep the tests honest

Data	Source	Coverage
FX spot, 7 currencies	FRED (Fed H.10)	1999–2026
3-month rates	OECD via FRED	1999–2026
Speculator positions	CFTC Commitments of Traders	1986–2026
Crypto price, volume, funding	Binance public archive	2020–2026
Carry factor (validation)	Lustig–Roussanov–Verdelhan	1983–2021

Data fix: before Aug 2000 CFTC lists the same currency futures under 'International Monetary Market'. Matching both names extended positioning back to 1986.

- All parameters were fixed before the first backtest.
- The out-of-sample period starts in 2011 for A and in 2024 for C.
- Costs are 3 bp per unit traded in FX and 5 bp in crypto.
- There is no look-ahead: weights set at t earn the return of $t+1$, and CFTC data are used only after their Friday release.
- There is no survivorship bias, because Terra (LUNA) and FTX Token (FTT) stay in the sample.
- Robustness grids are reported, but they were never used for tuning.

Strategy A: crowding-filtered G10 carry

Yen, euro, pound, Swiss franc, Canadian, Australian and New Zealand dollar vs USD, monthly

1

Carry portfolio

At each month-end we buy the 2 highest-rate currencies and sell the 2 lowest, with equal weights and no net dollar bet.

2

Crowding signal

Each currency's CFTC net speculative position over open interest is z-scored over 36 months.

3

Portfolio crowding

It is the average z of the long leg minus the average z of the short leg.

4

Filter

If crowding is above 1, all positions are cut to half for the next month.

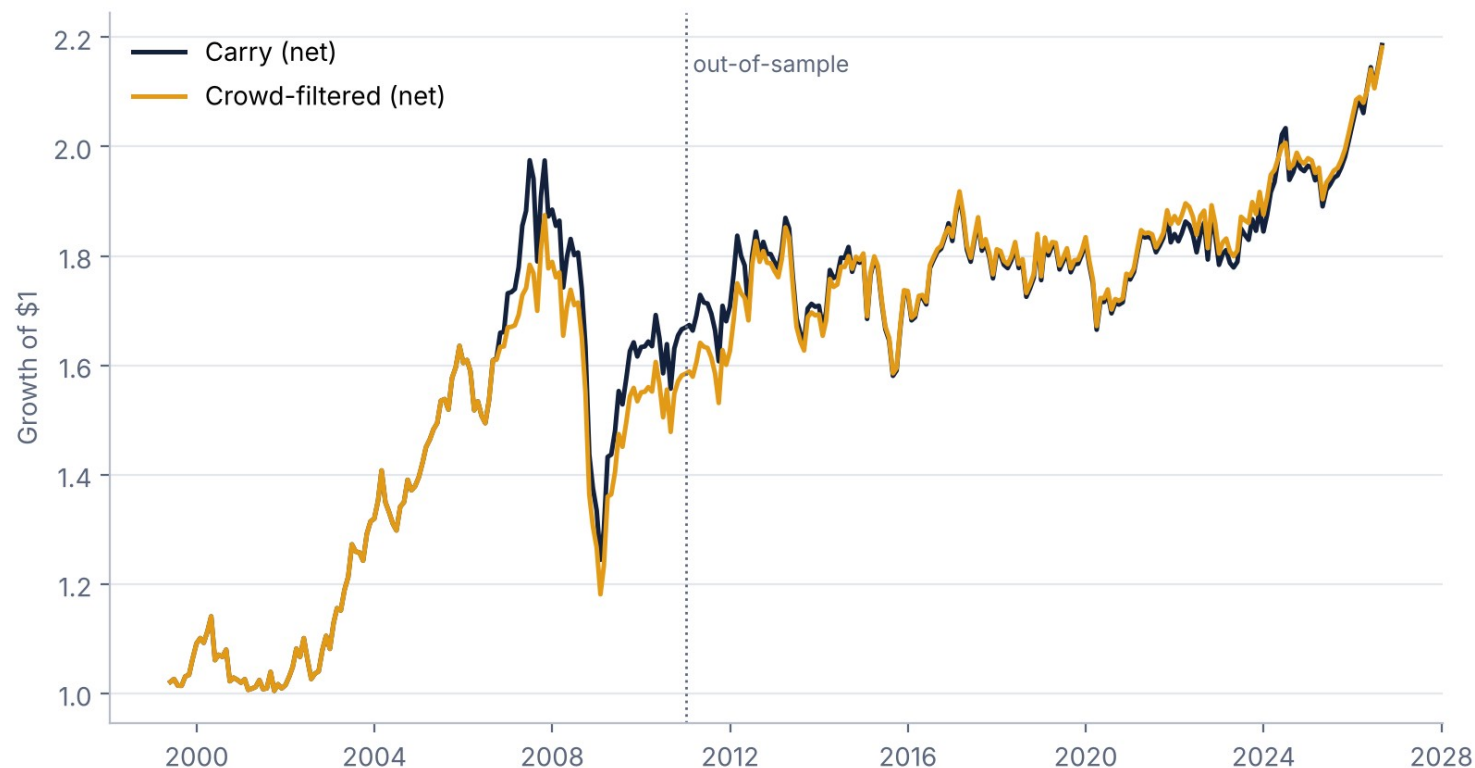
33

crowded months out of 328, 1999–2026

Who we usually hold

- Long: New Zealand dollar (94% of months) and Australian dollar (75%).
- Short: Swiss franc (99%), Japanese yen (52%) and euro (40%).

Crowding filter helps out of sample, not in 2008



0.27 → 0.32

Out-of-sample Sharpe, net: carry vs crowd-filtered (2011–2026)

0.30% vs 0.02%

Next-month carry after non-crowded vs crowded months

0.82

Correlation with the academic LRV carry factor (1999–2021)

Full sample 1999–2026, net of costs: carry / crowd-filtered

3.3% / 3.2%

Net return a year

8.9% / 8.5%

Volatility a year

-37% / -37%

Max drawdown

Strategy A loses when volatility spikes

Monthly net carry on risk factors, Newey-West t-stats, 1999–2021

-1.55

Beta to equity-volatility changes (t = -3.7)

0.35

Beta to the dollar factor (t = 4.1)

0.25%

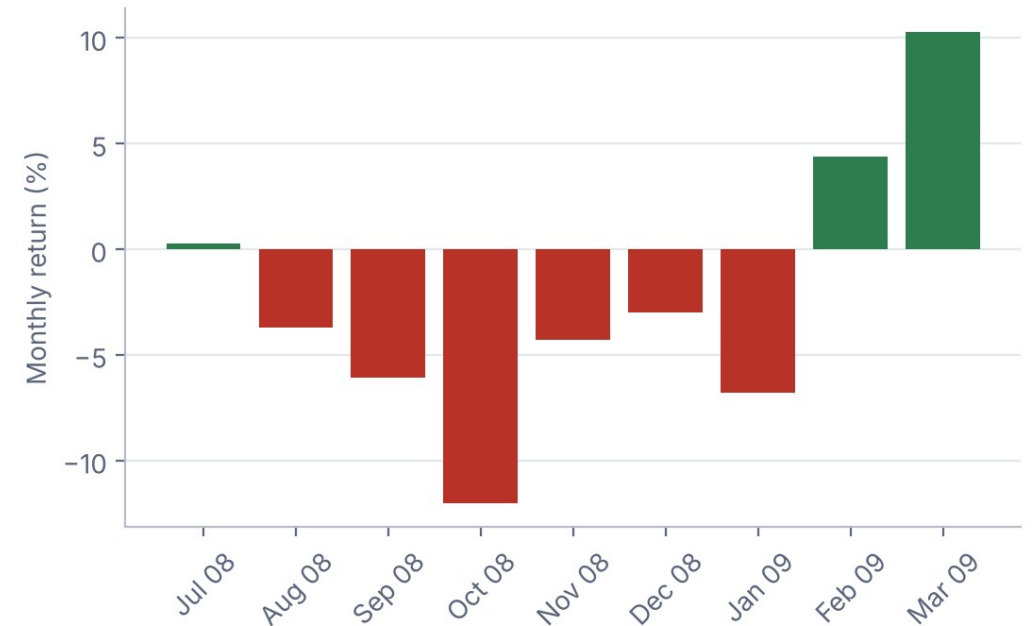
Monthly alpha (t = 1.7): not significant

0.18

R² of the two-factor model

Worst months: Oct 2008, Aug 2007, May 2000, Jan 2009, and the Jan 2015 Swiss franc de-peg.

The 2008 carry crash: -20.5% (Jul 2008 – Mar 2009)



Strategy C: crypto funding carry

Binance perpetuals on BTC, ETH, BNB, XRP, ADA, DOGE, SOL, LTC, LINK, AVAX, LUNA (Terra), FTX; weekly, 2020–2026



- Funding is paid every 8 hours, and positive funding means longs pay shorts.
- Each Sunday we short the third of coins with the highest 7-day funding and buy the lowest third.
- The book is dollar-neutral and pays 5 bp per trade.

0.14

Net Sharpe, full

0.81

Net Sharpe, 2024–26

Net of costs: full sample 2020–2026 / since 2024

4.1% / 13.9%

Net return a year

30% / 17%

Volatility a year

-67% / -13%

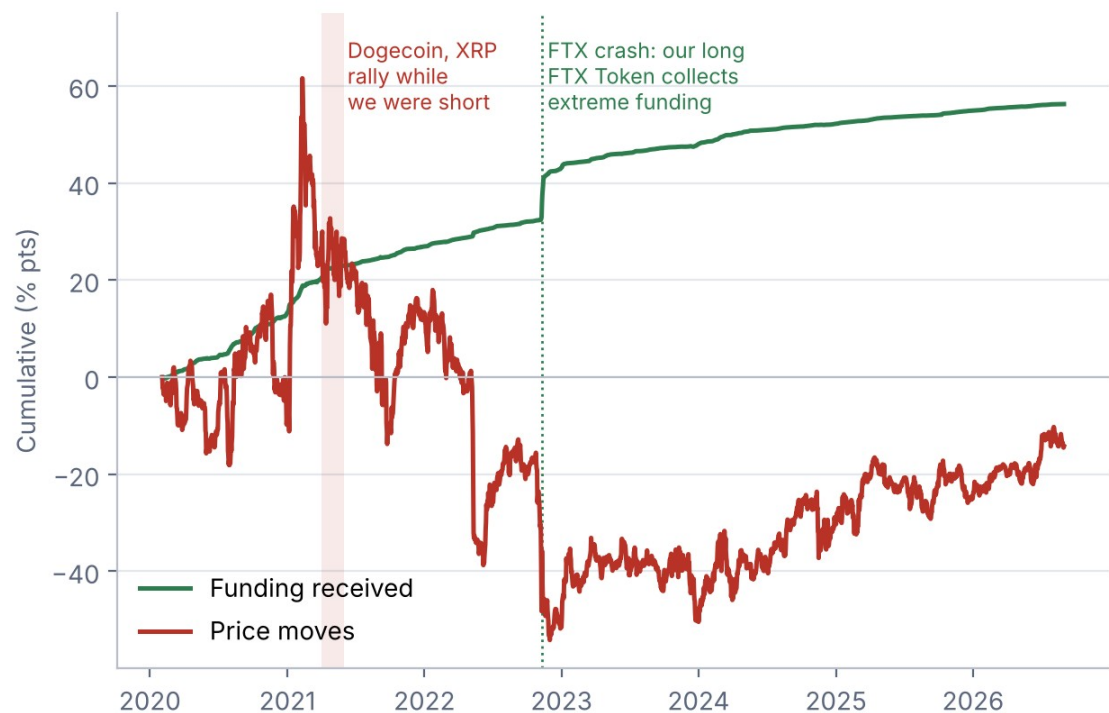
Max drawdown

-27%

Worst week: LUNA (Terra) collapse, May 2022

Funding is earned, but price moves dominate

Where the return comes from, and what drives the losses



Total P&L by coin, % pts: Solana +102, BNB +51, Avalanche +36; Terra -87, XRP -58, Dogecoin -43, FTX Token -13.

Robustness: all 6 lookback and rebalancing settings have a positive out-of-sample Sharpe (0.41 to 1.29); our pre-set choice is in the bottom half.

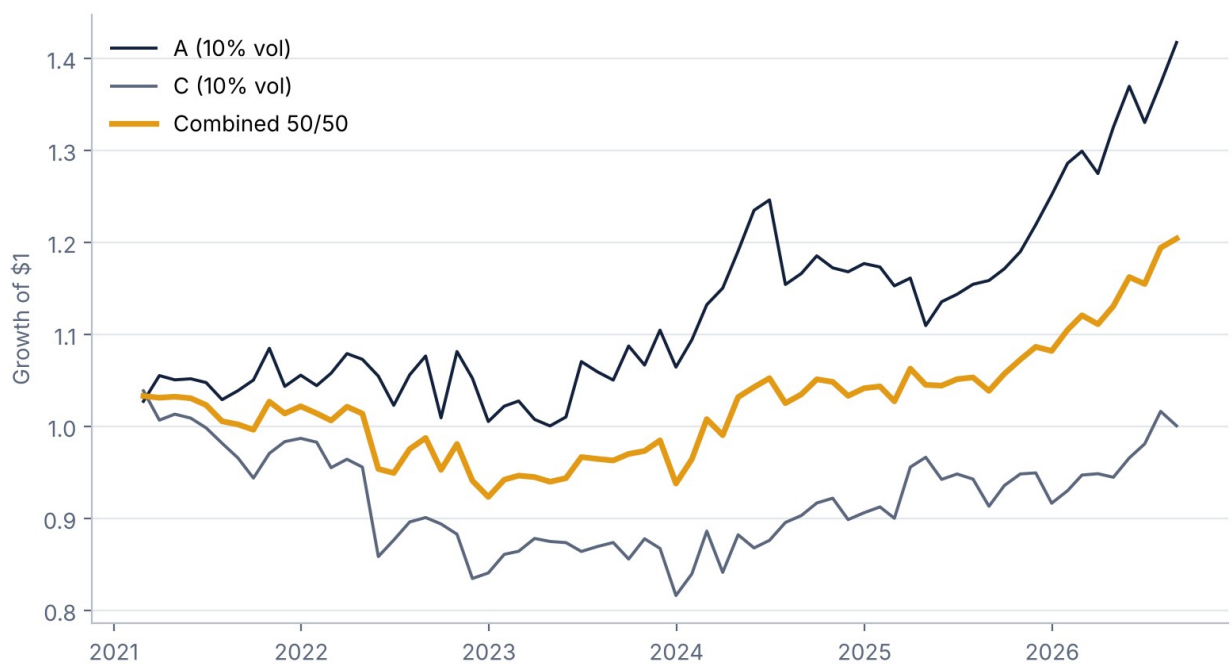
% per year	Funding	Price	Net
Full	+8.5	-2.2	+4.1
In-sample	+12.3	-12.5	-2.5
Out-of-sample	+3.1	+13.0	+13.9

Takeaways

- Funding income is steady at about 8.5% a year, but price moves decide the result.
- The strategy has no market exposure, since its beta to Bitcoin is -0.02.
- Its losses come from single coins: Terra lost 87 points, XRP 58 and Dogecoin 43.
- Our pre-set risk control lowered the Sharpe from 0.14 to -0.16, so it did not help.
- Deeply negative funding signals crowded shorts, not a cheap coin.

Combining FX and crypto carry

Each scaled to 10% vol with lagged volatility, 50/50, 2021-2026



	Sharpe	Vol (%)	Max DD (%)	Worst month (%)
A	0.71	9.4	-11	-7.4
C	0.05	9.3	-21	-10.2
Combined 50/50	0.51	7.0	-11	-5.9

0.11

Correlation between A and C, monthly, 2021-2026

Takeaways

- FX carry and crypto carry are almost uncorrelated, so they crash at different times.
- The 50/50 mix has 7.0% volatility, compared with about 9% for each strategy alone.
- Its worst month is -5.9%, smaller than for FX (-7.4%) or crypto (-10.2%).
- The mix does not beat FX alone on Sharpe (0.51 vs 0.71), because crypto carry was weak in this window.

Conclusions and limitations

Hypotheses: verdict and evidence

Hypothesis	Verdict	Evidence
A1: G10 carry earns a premium	Supported	Net Sharpe 0.37; 0.82 correlation with LRV
A2: crowding predicts weaker carry	Leaning yes	0.02% vs 0.30% next month; OOS Sharpe 0.27 → 0.32; missed 2008
A3: carry loses in volatility spikes	Supported	Vol beta -1.55 (t = -3.7); alpha not significant
C1: funding carry earns a premium	Mixed	Funding +8.5% a year, but net Sharpe 0.14; price moves dominate
C2: no BTC exposure; coin-specific crashes	Supported	BTC beta ≈ 0 ; Terra -87 pts; worst week -27%
Pre-set risk control helps	Not supported	Sharpe 0.14 → -0.16
FX and crypto carry diversify	Supported	Correlation 0.11; vol 7.0% vs about 9%

What is new

- We use CFTC speculator positioning as a crowding filter on G10 carry.
- We treat crypto funding as the crypto interest-rate gap and as a crowding signal, and test it with the coins that died.

Limitations

- The crypto history is short (2020–2026) and covers only 12 coins.
- There are only 33 crowded FX months, and the 2008 crash was not flagged.
- Costs are simple estimates, and we ignore funding-rate caps, borrow limits and exchange risk.